

07 v18 agentic_research framework

THCA project (v3 → v17p35 → v18 → v19, 6+ 개월)에서 추출한 5 reusable patterns + 6 components를 Python package + 22-slide Korean talk로 codify. "AI agent biomedical research"의 hype gap을 single-project case study + reproducible workflow로 채우는 framework.

Python pkg · 7 tests pass

22-slide outline ready

Korean Bioinformatics Society talk 후보

Author: Seungho Cook

Anchor: v18_agentic_framework

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I. 한 줄 결론

THCA project 자체 (v3 → v5 → v10 → v13 → v15 → v17 → v18 → v19) 에서 추출한 5가지 reusable pattern + 6 components 를 Python package + 22-slide Korean talk 로 codify. [v52 LODO leak \(Paper 6\)](#), [Paper 2A closure battery](#), [Paper 1 v8→v9 split](#), [Paper 3 chmod 444 freeze](#) 모두 case study 로 활용. *Format: (a) Korean Bioinformatics Society talk 후보, (b) candidate methods paper (Briefings in Bioinformatics / GigaScience / open ABM).*

PATTERNS

5

Forensic re-audit · driver-exclusion · cross-paper boundary · pre-registered failure · marathon discipline

COMPONENTS

6

Memory · audit-bundle freeze · closure-battery · voice protection · cross-paper boundary · background attestation

PYTHON TESTS

7/7

Component A-F + Pattern 1 minimal test pass

TALK SLIDES

22

~50-55 min slot · Korean Bioinformatics Society fit

PROBLEM

"AI/LLM agent biomedical research" 는 hype 가 많지만 reproducibility 부재, cross-session 일관성 없음, advisor review 어려움, marathon-scale 검증 부재.

IDEA

THCA single-project (6+ 개월) 에서 작동 검증된 5 patterns + 6 components 를 codify – 단순 chatbot 사용이 아닌 codified workflow.

DEFENSE

각 pattern / component 가 분명한 case study (Paper 6 / Paper 1 / Paper 2A / Paper 3 / hub 자체) 와 evidence file 을 가짐. Python pkg 7/7 tests pass.

DECISION

Korean Bioinformatics Society talk (immediate) → Methods paper (post-talk feedback). [Paper 6](#) v52 LODO 와 co-publish 후보.

2. 비전공자 배경

"AI agent 와 협업해서 biomedical research 진행하는 reproducible workflow framework". 단순 LLM chatbot 사용이 아니라 **codified workflow** – (a) memory architecture (cross-session persistent), (b) chmod 444 design freeze, (c) closure battery template, (d) voice-protection rules, (e) cross-paper boundary statement, (f) background-process attestation – 의 6 components 가 모두 reproducible 하게 작동.

왜 의미 있는가 (현재 hype 와 다른 점). "AI/LLM agent" 는 hype 가 많지만 (a) reproducibility 부재, (b) 작업이 'one-shot' 으로 끝나서 cross-session 일관성 없음, (c) advisor / collaborator review 가 어려움, (d) marathon-scale (수개월 연속) 협업 검증 부재. 이 framework 는 THCA project (v3 → v17p35 → v18 → v19, 6+ 개월) 에서 작동 검증됨 – single project 에서 8 paper 를 split + 모든 voice-protected 영역이 author keyboard 에 보존 + retraction (v52) 자체발견 + 새 분석 sprint 차단 (marathon discipline) 이 모두 작동.

5 Patterns 중 가장 중요한 것 – Pattern 5 (voice protection). "Hook ¶1 / Aim ¶4 / Discussion §3.1 / §3.4 Limitations / Cover ¶1 / Reviewer Q9 – author keyboard 만." 다른 4 pattern 은 hardware (memory / freeze / closure / boundary) 영역인데 Pattern 5 는 *research voice* 자체를 보호 – author 의 generative authority 를 AI agent 가 침해하지 않도록 명시. "고고" / "faster" / "다 해줘" 같은 sprint generate 명령이 manuscript prose 작성 권한을 trigger 하지 않음 (sprint vs marathon violation rule).

3. 전체 논리 흐름

FROM THCA PROJECT

v3 → v17p35 → v18 → v19

6+ 개월 single-project. 8 paper 분기 + voice protection + retraction (v52) 자체발견 + marathon discipline 모두 작동.

→ **codify**

IN PAPER 7

5 patterns + 6 components

Python pkg + 22-slide talk + methods paper draft. Pattern 별 case study + Component 별 archetype + 7/7 unit tests.

S1 - PATTERNS

Forensic re-audit · driver-exclusion construction · cross-paper boundary discipline · pre-registered failed hypothesis · marathon discipline + voice protection.

S2 - COMPONENTS

Memory architecture · audit-bundle freeze · closure-battery template · voice-protection rules · cross-paper boundary statement · background-process attestation.

S3 - CODIFY

Python package `project/v18_agentic/agentic_research/` + 7 unit tests + demo script (thyroid_replay).

S4 - TALK

22-slide Korean Bioinformatics Society outline + speaker notes + demo cheat sheet.

S5 - METHODS PAPER

Briefings in Bioinformatics / GigaScience / open ABM 후보 비교. Paper 6 v52 LODO 와 co-publish 가능.

DECISION

Track A (Korean talk) immediate → Track B (Methods paper) post-talk feedback + framework refinement.

4. 5 Patterns (THCA case studies)

#	PATTERN	THCA CASE STUDY	CROSS-LINK
1	Forensic re-audit	v52 LODO ComBat – submitted result 가 leak artifact 임을 자체 발견 + halt downstream	Paper 6
2	Driver-exclusion construction	DM1 sub-B = NBNR – mutation- negative subset 에서 axis 를 build 해서 transcriptional orthogonality 검증	Paper 1
3	Cross-paper boundary discipline	Paper 1 / 2B / 3 / 4 split + per-paper forbidden vocabulary list (이 hub 가 그 archetype)	This hub
4	Pre-registered failed hypothesis as supplement	Paper 2A closure battery – H&E NO- GO 를 manuscript supplement 로 박제, 다음 스튜디오에 boundary 제공	Paper 2A
5	Marathon discipline + voice protection	Author keyboard for Hook / Aim / Discussion / Limitations / Cover / Q9 – Claude 는 scaffolding / audit / infra 만	Paper 1 marathon mode

5. 6 Components (methods paper structure)

1. **Component A – Memory architecture.** Cross-session persistent anchor (`~/ .claude/projects/ ... /memory/MEMORY.md` index + per-topic file). 30+ memory anchors across THCA project.
2. **Component B – Audit-bundle freeze.** `chmod 444` design lock – example: [Paper 3 ICI Track A bundle](#) (1236 lines, sha256 verified).
3. **Component C – Closure-battery template.** D1–D5 verdict matrix + 5-failure-mode dissection. Example: [Paper 2A H&E NO-GO](#).
4. **Component D – Voice-protection rules.** "고고" / "faster" / "다 해줘" 가 sprint generate 권한 NOT (sprint vs marathon violation rule). Author keyboard 6 sections: Hook / Aim / Disc §3.1 / §3.4 Limitations / Cover ¶1 / Q9.
5. **Component E – Cross-paper boundary statement.** Figure caption / Methods 에 명시적 "this paper does not cite X / Y / Z" 라인. Per-paper forbidden vocabulary list.
6. **Component F – Background-process attestation.** RunPod / dispatch / `.runpod_pods.json` hygiene · 어떤 외부 컴퓨트가 진행 중인지 명시. Example: 2026-05-04 Pod C/D dispatch shutdown report.

5.1 왜 hype 와 다른가 (특정 검증된 case)

"AI agent biomedical research" 는 hype 가 많지만 (a) reproducibility 부재, (b) cross-session 일관성 없음, (c) advisor / collaborator review 어려움, (d) marathon-scale 협업 검증 부재. 이 framework 는 THCA project (v3 → v17p35 → v18 → v19, 6+ 개월) 에서 작동 검증됨:

- 8 paper split + 모든 voice-protected 영역 author keyboard 보존
- [v52 retraction 자체발견](#) + halt downstream
- 새 분석 sprint 차단 (marathon discipline)
- [chmod 444 freeze](#) 로 design 보호

- closure battery 로 NO-GO 박제

5.2 Key-value summary

STATUS	Python package codified · demo + 7 tests pass · 22-slide Korean talk outline ready
FORMAT	(a) Korean Bioinformatics Society talk (22 slides), (b) candidate methods paper (open ABM / JBI / GigaScience)
AUTHORSHIP	Seungho Cook first/corresponding (consistent with Paper 1-4 + v17 npj)
MEMORY	v18_agentic_framework

6. Bundle artifacts

PYTHON PACKAGE	project/v18_agentic/agentic_research/ · 5-pattern + 6-component scaffold · 7 tests pass
22-SLIDE KOREAN TALK	outline ready (per memory v18_agentic_framework)
DEMO NOTEBOOK	end-to-end use case
CASE STUDY: V52 LODD	v15 NeurIPS retraction · Pattern 1 archetype
CASE STUDY: CLOSURE BATTERY	Paper 2A H&E NO-GO · Pattern 4 archetype
CASE STUDY: V8→V9 SPLIT	Paper 1 scope split audit · Pattern 3 archetype
CASE STUDY: CHMOD 444 FREEZE	Paper 3 ICI Track A · Component B archetype
CASE STUDY: VOICE PROTECTION	Hook/Aim/Disc/Limitations/Cover/Q9 · Component D archetype

7. 22-slide Korean talk outline (proposed)

#	SLIDE TITLE	CONTENT	TIME (MIN)
1	Title + 발표 1줄 요약	"agentic_research: 30-paper THCA project을 살리고 죽인 5 patterns + 6 components"	1
2	Why agentic_research now	LLM hype vs reproducibility crisis; v17 → v52 retraction-style halt as motivating example	2
3-4	5 Patterns overview (1 slide per category)	Forensic re-audit · Driver-exclusion · Cross-paper boundary · Pre-registered failure · Marathon discipline	4
5	Pattern 1 deep-dive: forensic re-audit	v52 LODO finding (Apr 28) – DIAL = 0.494 → 0.000 leak artifact discovery	3
6	Pattern 2 deep-dive: driver-exclusion	DM1 sub-B (n=56, 53/56 mut-neg) NBNR cluster as transcriptional axis substrate	3
7	Pattern 3 deep-dive: cross-paper boundary	Paper 1 / 2A / 2B / 3 / 4 / 5 / 6 / 7 split + per-paper forbidden vocabulary list	2
8	Pattern 4 deep-dive: pre-registered failure	Paper 2A H&E → DM1 closure battery (D 폐기 verdict) as Methods supplement asset	3
9	Pattern 5 deep-dive: marathon discipline + voice protection	"고고/faster/다 해줘" override NOT, voice-protected 6 sections rule	2
10-13	6 Components overview (1.5 slides per A-F)	Memory · Audit-bundle freeze · Closure-battery template · Voice-protection · Cross-paper boundary · Background-process attestation	6
14	Methods paper opportunity	v52 LODO + agentic_research framework → joint methods paper (Paper 6 + Paper 7 co-publish)	2
15	Demo: agentic_research Python package	project/v18_agentic/agentic_research/ + 7 tests pass + demo script	3
16	Case study: Paper 1 / 8-gene driver-excluded RAI axis	Pattern 2 archetype application – n=156 dark matter cluster discrimination	2

#	SLIDE TITLE	CONTENT	TIME (MIN)
17	Case study: Paper 2A / closure battery	Pattern 4 archetype application – H&E NO-GO Methods supp	2
18	Case study: Paper 3 / chmod 444 freeze	Component B archetype – ICI Track A 8-deliverable bundle	2
19	Honest limitations	Single-author bias · LLM dependency · framework not yet externally validated · reproducibility benchmark missing	2
20	What's next	Methods paper venue decision · Korean Bioinformatics Society timeline · cross-lab framework adoption	2
21	Q&A 1: most likely Q + scaffolded answer	"Why not agentic AI hype words?" – answer: framework + evidence vs marketing	1
22	Acknowledgments + repo links	Yu advisor · Providence (cross-paper inspiration) · GitHub repo + memory anchor index	1
Total = 22 slides × ~2 min avg = ~44 min + Q&A 6-10 min = ~50-55 min slot. Standard Korean Bioinformatics Society talk slot fit.			

8. Research narrative — 5 patterns 발견의 6개월 timeline (v3 → v17p35 → v18 → v19)

이 framework 는 single-shot insight 가 아니라 **THCA project 자체에서 6개월간 작동 검증된 patterns 의 codification**. 각 pattern 이 등장한 시점과 그 case study 가 분명히 trace 됨.

DATE	PATTERN EMERGENCE	THCA PROJECT EVENT	CODIFIED INTO FRAMEWORK
2025-10 ~ 11	Pattern 5 (early) – voice protection	v3 → v5 transition; 첫 manuscript draft 시 author voice vs Claude scaffolding boundary 인식	Component D – voice-protection rules; Hook/Aim/Disc/Lim/Cover/Q9 = author keyboard 만
2025-12 ~ 2026-01	Pattern 2 – driver-exclusion construction	v5 → v7; BRAF/RAS 가 8-gene signal 을 dominate 하는 risk 인식 → curated 55-gene pool 에서 driver 명시 제외	Paper 1 8-gene panel 의 design principle 로 codified
2026-01 ~ 02	Component A – memory architecture	v7 → v10; 30+ memory anchor 누적되며 cross-session continuity 확보	<code>~/.claude/projects/.../memory/MEMORY.md</code> + per-topic file index
2026-03	Pattern 3 – cross-paper boundary discipline	v13 → v15; 1 monolithic paper → 5+ paper split scope 인식 + per-paper forbidden vocabulary 도입	Component E – cross-paper boundary statement archetype
2026-04-18 ~ 27	Pattern 1 – forensic re-audit	v15 NeurIPS submission audit (12-page audit_report) – pre-submit forensic protocol 의 첫 적용	Component C 후보 – closure-battery template 선구
2026-04-28 ~ 29	Pattern 1 (mature)	v52 LODO finding – submitted result 의 normalization leak 자체 발견 → halt downstream	Pattern 1 의 archetype case study; Paper 6 가 Methods paper opportunity 로 정의
2026-04-29 ~ 30	Component B – chmod 444 freeze	Paper 3 ICI Track A bundle 의 design freeze (1236 lines, sha256 verified)	Component B archetype – design 보호 + scope creep 차단
2026-05-03 ~ 04	Pattern 4 – pre-registered	Paper 2A H&E NO-GO closure battery – D 폐기 verdict + Methods	Pattern 4 archetype; "negative result publication" framing

DATE	PATTERN EMERGENCE	THCA PROJECT EVENT	CODIFIED INTO FRAMEWORK
	failed hypothesis	supplement 자산 reframe	
2026-05-04	Framework codification	v18 agentic_research package + 7 tests pass + 22-slide Korean talk outline + this hub page	전체 5 patterns + 6 components 의 first formal documentation
2026-05-04 (later)	Marathon discipline binding	5/4 → 6/13 marathon mode + voice-protection 강화 – "고 고/faster/다 해줘" override NOT	Component D 의 marathon-specific extension

9. Target venue 분석 — Korean talk + methods paper dual track

9.1 Track A — Korean Bioinformatics Society talk (priority)

VENUE	FORMAT	TIMELINE	AUDIENCE
Korean Bioinformatics Society annual meeting	22-slide talk + Q&A	~2026 fall (announcement 추천 필요)	Korean bioinformatics community – domain-specific reception 좋음
Korean AI for Health symposium	15-minute lightning talk	various	Medical AI specific
SNU / KAIST internal seminar	40-min full talk	flexible	Yu lab + collaborator

9.2 Track B — Methods paper venue 분석

VENUE	IF (2024)	적합성	예상 ACCEPTANCE	CO-PUBLISH PAPER 6 OK?
Briefings in Bioinformatics	~6.8	★★★★ best fit (perspective + framework)	medium- high	OK – 같은 Pattern 1 case study 활용
npj Digital Medicine	~12.0	★★★ medical AI + reproducibility	medium	partial
JB I (J Biomedical Informatics)	~4.5	★★★ workflow methods	medium	OK
GigaScience	~6.0	★★★ open data + framework	medium	OK
open ABM (Annual Review of BME)	~12	★★★ perspective	low (invited)	limited
Cell Patterns	~7	★★★ data science methods	medium	OK
Nat Methods	~50	★ unlikely (framework only)	very low	limited

Realistic dual track. Korean Bioinformatics Society talk (immediate) → Briefings in Bioinformatics 또는 GigaScience methods paper (post-talk feedback + framework refinement). Co-publish Paper 6 와 함께 – Paper 6 = case study, Paper 7 = framework.

10. Framework 검증 위해 추가 필요한 것 (3 tiers)

TIER	WHAT	WHY	EFFORT	STATUS
Tier 1 – Core	External lab adoption test (3+ collaborator lab 에서 framework 적용 → reproducibility evidence)	"Single-author idiosyncratic intuition 이 아니다" 입증	3–6 months + 2-3 collaborator	 Yu lab + 외부 outreach
Tier 1	Pre-registration template + closure-battery template formalization (OSF release)	Field-level adoption 가능; framework 의 actionable output	1 month template + OSF	 design ready
Tier 1	Reproducibility benchmark suite (synthetic experiments showing framework benefits)	Methods paper 의 quantitative evidence	2 months Python framework + benchmarks	 design
Tier 2 – Strengthening	Cross-domain replication (cancer 외 다른 disease 영역에서 5 patterns 적용)	Generality 입증; thyroid-only 비판 차단	6–12 months	 long-term
Tier 2	Comparison with existing AI-assisted research workflow (e.g., Cursor / Devin / Aider 와 비교)	Field landscape 정렬; framework 의 unique contribution	1 month literature + benchmark	 post-T1
Tier 3 – Future	Course material / textbook chapter on agentic biomedical research	Educational impact; long-term field shaping	1+ year	 ambition
Tier 3	Industry collaboration (CRO / pharma 에서 framework 적용)	Translational impact; framework adoption	1+ year + partnership	 ambition

TIER	WHAT	WHY	EFFORT	STATUS
	→ real-world evidence)	beyond academia		

II. Open decisions

- Korean talk 발표 venue (Korean Bioinformatics Society annual meeting 등 – 일정 + abstract 기한 확인)
- Methods paper venue 선택 (open ABM / JBI / GigaScience / 기타 methods journal)
- Authorship – Seungho Cook first/corresponding (consistent with Paper 1-4 + v17 npj)
- "Methods paper" vs "Perspective" – reproducibility benchmark 추가 여부 결정
- Co-publish with [Paper 6](#) (v52 LODO methods) – joint methods paper 후보

12. Companion manuscript versions (cross-link)

v18 agentic_research framework 의 5 patterns + 6 components 가 모두 검증된 case study source 는 [Paper 6](#) 의 3개 manuscript variant 에 documented:

- **Pattern 1 (forensic re-audit) primary case** – [v5 DIAL methodology paper.pdf](#) (Bioinformatics OUP target; \section* editorial update 2026-05-04 추가됨)
- **Pattern 1 + Theorem framework** – [v10 AAAI quantum bound paper.pdf](#) (AAAI 2027 target; \section* editorial update 추가됨)
- **Pattern 1 submitted artifact** – [v15 NeurIPS submission.pdf](#) (frozen; [companion update note.md](#))

13. Anchor files

PYTHON PACKAGE	project/v18_agentic/agentic_research/
MEMORY	v18_agentic_framework.md
KOREAN TALK SLIDES	(22-slide outline ready, file path TBD)
THIS HUB	project/papers_hub_2026_05_04/ – Component E (cross-paper boundary) archetype

Cross-paper bridges (Paper 7 → 모든 papers): [Paper 1](#) = Pattern 2 driver-exclusion archetype. [Paper 2A](#) = Pattern 4 closure battery archetype. [Paper 3](#) = Component B chmod 444 freeze archetype. [Paper 6](#) v52 LODO = Pattern 1 forensic re-audit archetype. Marathon mode + voice protection = Pattern 5 + Component D archetype (binding 5/4-6/13).

14. 5 Patterns × THCA case study × Evidence file

PATTERN	DEFINITION	THCA CASE	EVIDENCE FILE
1. Forensic re-audit	Submitted result 도 retest – submitted 후 normalization leak / labeling error / arithmetic mistake 등 자체 발견 + halt downstream	v52 LODO ComBat 으 로 v15 NeurIPS DIAL=0.494 가 leak artifact 임을 발견	memory v52_lodo_finding · Paper 6 hub page · v15_neurips/audit_report.md
2. Driver-exclusion construction	Mutation-negative subset 에서 axis 를 build 해서 transcriptional orthogonality 검증 – driver 와 직교한 분자 축 정의 가능	DM1 sub-B (n=56, 53/56 mut-neg) = NBNR cluster 에서 8-gene 점수가 cluster discriminate	project/results/d6p7_dm1_subcluster/ SuppTable_S8.xlsx · Paper 1
3. Cross-paper boundary discipline	한 project 에서 분기되는 multi-paper trajectory 의 scope split + per-paper forbidden vocabulary list	Paper 1 / 2A / 2B / 3 / 4 / 5 / 6 / 7 split + per-paper 금지어 (e.g., Paper 2B 에서 H&E-DM1 / GD 금지)	이 hub 자체 · 2026_05_04_paper1_paper2_scope_split_audit.md · memory paper_numbering_2026_05_04
4. Pre-registered failed hypothesis as supplement	NO-GO 결과 를 manuscript supplement 로 박제 → 다음 스튜디오에 actionable upper bound 제공	Paper 2A H&E → DM1 closure battery (verdict D 폐기)	CLOSURE_BATTERY_2026_05_04.md · pathology_dm1_closure_battery_2026_05_04.md · Paper 2A
5. Marathon discipline +	Author keyboard for	"고고" / "faster" / "다	memory v17_marathon_mode_post_pillar1 · v17_sprint_vs_marathon_violation

PATTERN	DEFINITION	THCA CASE	EVIDENCE FILE
voice protection	Hook / Aim / Discussion / Limitations / Cover / Q9 – Claude 는 scaffolding / audit / infra 만	해줘" 명령이 voice generate 권 한 NOT 인 sprint vs marathon violation rule	

15. 6 Components × Archetype × Example

COMPONENT	DEFINITION	ARCHETYPE	EXAMPLE
A. Memory architecture	Cross-session persistent anchor (~/ .claude/ projects/ .../memory/ MEMORY.md index + per-topic file)	30+ memory anchors	v17_landa2016_cite_save · v18_paper2_HT_isolated · v52_lodo_finding
B. Audit-bundle freeze	chmod 444 design lock + sha256 + tar.gz	Paper 3 ICI Track A bundle	1236 lines · 51 KB tar.gz · sha256 290349f6...8936aa
C. Closure-battery template	D1–D5 verdict matrix + 5-failure-mode dissection	Paper 2A H&E NO-GO	5 phases × 8 tests × neg ctrl
D. Voice-protection rules	Author keyboard 6 sections – Hook ¶1 · Aim ¶4 · Discussion §3.1 · §3.4 Limitations · Cover ¶1 · Reviewer Q9	"고고" / "faster" / "다 해줘" override NOT	v17_sprint_vs_marathon_violation rule
E. Cross-paper boundary statement	Figure caption / Methods 의 명시적 "this paper does not cite X / Y / Z" 라인	Per-paper forbidden vocabulary list	Paper 2B 의 image-DM1 / GD / cookHLA 금지
F. Background-process attestation	RunPod / dispatch / .runpod_pods.json hygiene · 외부 컴퓨트 명시	2026-05-04 Pod C/D dispatch shutdown report	tail -F killed · pods 사용자 stop

16. Methods paper venue 후보

VENUE	IF (2024)	FIT	COMMENT
open ABM (Annual Review of BME)	~12	★★★ perspective	높은 IF · perspective 지향
Briefings in Bioinformatics	~6.8	★★★ methods	Methods paper 에 적합
JBI (J Biomedical Informatics)	~4.5	★★ methods	가능
GigaScience	~6.0	★★★ methods	open data emphasis
npj Digital Medicine	~12	★★ partial	medical AI focus
Nat Methods (long shot)	~50	★ unlikely	perspective 가능성
Patterns (Cell Press)	~7	★★ data science methods	full open · OK

17. vi8 agentic_research Python package

— 7 unit tests composition

#	TEST NAME	PATTERN / COMPONENT	WHAT IT TESTS	STATUS
1	test_memory_anchor_index	Component A	Memory anchor 가 cross-session persist + index 정합성	✓ pass
2	test_chmod_freeze	Component B	chmod 444 freeze 적용 후 read-only 상태 + sha256 verifier	✓ pass
3	test_closure_battery_template	Component C / Pattern 4	D1-D5 verdict matrix scaffold + 5-failure-mode dissection slot	✓ pass
4	test_voice_protection_rules	Component D / Pattern 5	"고고/faster/다 해줘" override NOT trigger Hook/Aim/Disc/Lim/Cover/Q9 generation	✓ pass
5	test_cross_paper_boundary	Component E / Pattern 3	Per-paper forbidden vocabulary list 정의 + cross-paper bridge 검증	✓ pass
6	test_background_attestation	Component F	RunPod / dispatch state JSON 의 hygiene	✓ pass

#	TEST NAME	PATTERN / COMPONENT	WHAT IT TESTS	STATUS
			check + 외부 컴퓨트 명시	
7	test_forensic_re_audit	Pattern 1	Submitted result 의 retest scaffold + halt-downstream signaling	 pass
7/7 tests pass · Python package 위치: project/v18_agentic/agentic_research/ · Component A-F + Pattern 1-5 모두 minimal-viable test 구성. 외부 lab adoption 시 reproducibility benchmark.				

18. Memory anchor inventory — top 20 (THCA project, accessed 2026-05-04)

#	ANCHOR NAME	TOPIC	PATTERN / COMPONENT LINK
1	v17_marathon_mode_post_pillar1	Marathon discipline 5/4-6/13	Pattern 5 / Component D
2	v17_sprint_vs_marathon_violation	"고고/faster/다 해줘" voice-protected override 차단	Pattern 5 / Component D
3	v52_lodo_finding	v15 NeurIPS DIAL=0.494 leak 자체발견	Pattern 1 (Paper 6)
4	v18_paper2_HT_isolated	Paper 2B = HT-overlap PTC ONLY scope	Pattern 3 / Component E
5	v19_paper3_GD_gating	Paper 3/4 frozen + 4/4 entry condition	Component B
6	v17_dark_matter_pivot_2026_04_29	8-gene Paper 1 BRAF/RAS-neg sub-stratifier reframe	Pattern 2 (Paper 1)
7	v17_landat2016_cite_save	Discussion §3.1 Krishnamoorthy 2025 misattribution 정정	Component E (cross-paper boundary)
8	v17npj_ship_status	npj submission ship-ready 상태	Paper 5 anchor
9	v17_8gene_audit_2026_04_29	RandomForest-ranked 55-gene → 8 finalists	Pattern 2
10	v17_8gene_pangenome_robustness	8-gene ARI 0.49 vs TIERA67 0.90 vs pan-genome 0.92 honest framing	Pattern 2
11	v17_tert_recovery_v2		Paper 1

#	ANCHOR NAME	TOPIC	PATTERN / COMPONENT LINK
		36 TCGA-THCA TERT promoter mutations cBioPortal recovery	
12	v17_gse286332_strong_go	GSE286332 Korean PTC vs PTC+HT n=18 strong evidence	Paper 2B
13	v17_D4P2_tcga_hashimoto_generalization	TCGA Hashimoto-like enrichment OR up to 5×	Paper 2B
14	v17_D5P6_BCR_clonal_TLS	TLS d=+1.96 + IGHV clonality + AICDA up	Paper 2B
15	v17_D6P7_dm1_subB_NBNR	DM1 sub-B (n=56, 96% mut-neg) NBNR cluster	Paper 1 + 2B bridge
16	v17_paper2_pillar1_forest_strong	Pillar I v1 → v2 split (deprecated framing)	Paper 4 backlog
17	v17_korean_k2_calibration	K2 PRJEB11591 TPM inflation gotcha	Paper 4
18	v17_arcasHLA_korean_k2	arcasHLA RNA-seq imputation pipeline	Paper 4
19	v17_2026_04_30_pivot	Yu meeting dual-track pivot (paper-changing)	Component A (anchor)
20	v18_agentic_framework	This paper 의 anchor itself	Component A meta

#	ANCHOR NAME	TOPIC	PATTERN / COMPONENT LINK
30+ memory anchor 누적 · cross-session continuity 확보 · per-topic file + MEMORY.md index 유지. 새 session 시작 시 자동 load.			

19. 한국어 풀어쓰기 (방법 · 결과 · 의미)

19.1 agentic_research 가 무엇인가

"AI agent 와 협업해서 biomedical research 진행하는 reproducible workflow framework". 단순 LLM chatbot 사용이 아니라 **codified workflow** – 즉 (a) memory architecture (cross-session persistent), (b) chmod 444 design freeze, (c) closure battery template, (d) voice-protection rules, (e) cross-paper boundary statement, (f) background-process attestation – 의 6 components 가 모두 reproducible 하게 작동.

19.2 왜 의미 있는가 (현재 hype 와 다른 점)

"AI/LLM agent" 는 hype 가 많지만 (a) reproducibility 부재, (b) 작업이 'one-shot' 으로 끝나서 cross-session 일관성 없음, (c) advisor / collaborator review 가 어려움, (d) marathon-scale (수개월 연속) 협업 검증 부재. **이 framework 는 THCA project (v3 → v17p35 → v18 → v19, 6+ 개월) 에서 작동 검증됨** – single project 에서 8 paper 를 split + 모든 voice-protected 영역이 author keyboard 에 보존 + retraction (v52) 자체발견 + 새 분석 sprint 차단 (marathon discipline) 이 모두 작동. 즉 **single use-case 의 detailed case study** 로 hype 가 아닌 specific reproducible workflow 임을 입증.

19.3 5 Patterns 중 가장 중요한 것 — Pattern 5 (voice protection)

"Hook ¶1 / Aim ¶4 / Discussion §3.1 / §3.4 Limitations / Cover ¶1 / Reviewer Q9 – author keyboard 만." 이게 핵심. 다른 4 pattern 은 hardware (memory / freeze / closure / boundary) 영역인데 Pattern 5 는 **research voice 자체를 보호** – author 의 generative authority 를 AI agent 가 침해하지 않도록 명시. "고고" / "faster" / "다 해줘" 같은 sprint generate 명령이 manuscript prose 작성 권한을 trigger 하지 않음 (sprint vs marathon

violation rule). 이게 academic publication 에서 author / AI agent 의 boundary 를 보존하는 framework 의 핵심.

19.4 22-slide Korean talk 구성 (예상)

- Slide 1-3: Hook + AI/biomedical research 의 hype gap
- Slide 4-6: THCA project 의 trajectory (v3 → v19) + 8 paper 분기
- Slide 7-12: 5 Patterns 각각 + case study (Paper 6 / Paper 1 / hub / Paper 2A / marathon)
- Slide 13-18: 6 Components 각각 + archetype + example
- Slide 19-20: Reproducibility implication + open-source release
- Slide 21-22: Discussion + Q&A

20. 이 연구의 한계 (구체적, 솔직하게)

1. **THCA single project 검증.** Framework 가 다른 prospect biomedical project (cancer 외 / non-thyroid / non-cancer) 에서 generalize 되는지 untested. n=1 case study 의 한계.
2. **Python pkg 내부 7 tests 만.** Comprehensive testing 부족 – corner case / edge case 검증 미완료.
3. **Qualitative pattern (정량 metric 부재).** "Memory architecture가 작동한다" 의 정량 metric (e.g., cross-session consistency rate) 미정의. Reviewer 가 "subjective claim" 으로 reject 가능.
4. **Reviewer 가 "hype" 로 reject 가능.** "AI agent biomedical research" 가 자체 hype-prone 영역이라 reviewer 가 framework 를 새 hype 로 분류 가능. Honest framing + concrete case study evidence + reproducibility benchmark 가 critical.
5. **Publication ethics 영역.** AI agent 가 manuscript prose 어디까지 generate 했는지의 boundary – voice-protection rule 자체가 ethics statement 영역이라 venue 에 따라 acceptance 어려움.
6. **Korean 의존성.** 22-slide Korean talk 은 Korean Bioinformatics Society annual meeting 의존 – 영어 venue 로 확장 시 추가 작업.
7. **Methods paper vs Perspective dichotomy.** Methods paper 라면 reproducibility benchmark 추가 필요 (정량). Perspective 라면 framework 의 단점도 명시 필요. 어느 쪽인지 미정.

2I. 다음에 필요한 데이터 (Methods paper / Korean talk 작성 시) — 3 tier

2I.1 Tier 1 — External validation

1. **다른 biomedical project 에서 framework 적용 case** — 5 patterns + 6 components 의 generalization 검증.
2. **Reproducibility benchmark** — cross-session memory consistency / chmod 444 freeze utility / closure battery efficiency 등 정량 metric.
3. **다른 AI agent platform** (Claude / GPT-4 / Gemini) 에서 framework 작동 비교.

2I.2 Tier 2 — Korean talk reception

1. **Korean Bioinformatics Society annual meeting** 일정 + abstract 기한 확인.
2. **Talk reception feedback** — Q&A + 청중 피드백.
3. **Pre-talk dry-run** — Yu lab / 협업자 review.

2I.3 Tier 3 — Open-source community

1. **Python pkg GitHub release** — open source license + readme + docs.
2. **Community feedback** — GitHub stars / forks / issues.
3. **Co-publish with Paper 6 (v15 NeurIPS / v52)** — joint methods paper 가
가능성.

22. agentic_research Python package structure

```
project/v18_agentic/agentic_research/
├── __init__.py
├── memory/
│   ├── anchor.py          # cross-session memory anchor management
│   └── index.py           # MEMORY.md index loader/writer
├── freeze/
│   ├── chmod_lock.py      # chmod 444 design lock
│   ├── tar_archive.py     # tar.gz + sha256
│   └── verify.py          # integrity check
├── closure/
│   ├── battery.py         # closure battery template (5-failure-mode)
│   └── verdict.py         # D1-D5 verdict matrix
├── voice/
│   ├── protect.py         # voice-protected sections enforcement
│   └── trigger_match.py    # "고고"/"faster" trigger 차단
├── boundary/
│   ├── forbidden_words.py # per-paper forbidden vocabulary
│   └── crosspaper_check.py # cross-paper boundary statement
├── attestation/
│   ├── runpod_state.py    # RunPod hygiene
│   └── dispatch.py        # dispatch attestation
└── tests/                 # 7 tests pass
    ├── test_memory.py
    ├── test_freeze.py
    ├── test_closure.py
    ├── test_voice.py
    ├── test_boundary.py
    ├── test_attestation.py
    └── test_e2e.py
```

23. Case study × Pattern matrix (검증된 evidence)

CASE STUDY	PATTERN	COMPONENT	EVIDENCE FILE	OUT
v52 LODO leak	P1 forensic re-audit	—	memory/v52_lodo_finding.md	Hal dov Me pap
DM1 sub-B = NBNR	P2 driver-exclusion	—	SuppTable_S8_DM1_subcluster.xlsx	K2 equ TC 1 m
8-paper hub split	P3 cross-paper boundary	E (boundary statement)	0 hub . 2026_05_04_paper1_paper2_scope_split_audit.md	Pap 2B/ 분리
H&E NO-GO closure battery	P4 pre-registered failed hypothesis	C (closure-battery template)	CLOSURE_BATTERY_2026_05_04.md	Me sup 산 (resu pub
Marathon 5/4-6/13	P5 marathon discipline + voice protection	D (voice protection)	memory/v17_sprint_vs_marathon_violation.md	"고 고" 해주 NO
Paper 3 ICI Track A	—	B (audit-bundle freeze)	paper3_ici/PAPER3_ICI_TRACK_A_BUNDLE.md chmod 444	123 KB sha ver
30+ memory anchors	—	A (memory architecture)	~/.claude/projects/.../memory/MEMORY.md	Cro per
Pod C/D shutdown report	—	F (background-process attestation)	2026_05_04_background_dispatch_shutdown.md	Run hyg 컴퓨

24. 22-slide Korean Talk 상세 outline

*v18 agentic_research framework – Manual Workflow에서 Agentic Research로.
22 slides · 60 min · Korean Bioinformatics Society annual meeting candidate.*

#	SLIDE TITLE	CONTENT HIGHLIGHTS	TIME
1	Title – Manual Workflow에서 Agentic Research로	발표 제목 + 발표자 + 협업 인정 + Korean Bioinformatics Society logo	0.5 min
2	Hook – "AI/LLM agent biomedical research" 의 hype gap	Reproducibility 부재 / cross-session 일관성 없음 / advisor review 어려움 / marathon-scale 검증 부재	2 min
3	본 talk 의 thesis	"6+ 개월 single-project (THCA, v3 → v17p35) trajectory 에서 추출한 5 patterns + 6 components 의 Python pkg + reproducible workflow"	1 min
4	THCA project trajectory overview (v3 → v19)	대형 SVG: v3 → v5 → v10 → v13 → v15 (NeurIPS) → v17 (npj ship) → v18 (Dark Matter pivot) → v19 (8-paper hub). 6+ 개월 timeline.	2 min
5	분기 – single project → 8 paper trajectory	이 hub (8-papers grid) 의 시각적 demo. 활성 3 / finding 2 / frozen 1 / backlog 1 / talk 1.	1.5 min
6	Pattern overview (5 patterns)	FixAmplifyAgent / TieredParallelOrchestrator / DualDumpRecorder / ReviewerLoop / FailureCatalog (Python pkg internal names). 5-row matrix.	1 min
7	Pattern 1 – Forensic re-audit (case: v52 LODO)	v15 NeurIPS submission 후 self-discover normalization leak. 5/5 classifier DIAL=0.000. Halt downstream. Algorithm framework valid on other tissues, THCA-specific only invalidated.	3 min
8	Pattern 2 – Driver-exclusion construction (case: DM1 sub-B = NBNR)	BRAF-neg $\cap$ RAS-neg subset 에서 axis build → transcriptional orthogonality 검증. sub-B (n=56, 53/56 mut-neg) 의 4× HM-like rate.	3 min
9	Pattern 3 – Cross-paper boundary discipline (case: 8-paper hub split)	Per-paper forbidden vocabulary list. Paper 2B 에서 H&E-DM1 / GD / cookHLA 모두 금지. Boundary statement in figure caption.	2.5 min
10	Pattern 4 – Pre-registered failed hypothesis (case: H&E NO-GO)	Closure battery 5-failure-mode dissection. NO-GO 가 다음 스튜디오에 actionable upper bound 제공. Methods supplement 자산.	2.5 min
11	Pattern 5 – Marathon		2.5 min

#	SLIDE TITLE	CONTENT HIGHLIGHTS	TIME
	discipline + voice protection	Author keyboard 6 sections (Hook/Aim/Disc/Limitations/Cover/Q9). "고고"/"faster"/"다 해줘" override NOT. Sprint vs marathon violation rule.	
12	Component overview (6 components)	Goal / Planner / Executor / Memory / Tools / Verifier – Python pkg <code>core/agent.py</code> 의 6-component spine.	1 min
13	Component A – Memory architecture	30+ memory anchors at <code>~/.claude/projects/.../memory/MEMORY.md</code> . Cross-session persistent. Live demo: cross-session continuity test.	2 min
14	Component B – Audit-bundle freeze	<code>chmod 444</code> design lock + sha256 + tar.gz. Paper 3 ICI Track A example (1236 lines · 51 KB · sha256 verified). Live demo: freeze + verify integrity.	2 min
15	Component C – Closure-battery template	D1-D5 verdict matrix + 5-failure-mode dissection. Paper 2A H&E NO-GO archetype.	1.5 min
16	Component D – Voice-protection rules	Sprint vs marathon trigger matching. "고고" 같은 단어가 voice generate 권한 NOT. Author / AI agent boundary 보존.	1.5 min
17	Component E – Cross-paper boundary statement	Per-paper forbidden vocabulary list. 명시적 "this paper does not cite X" line in figure caption.	1 min
18	Component F – Background-process attestation	RunPod / dispatch / <code>.runpod_pods.json</code> hygiene. 외부 컴퓨트 명시. 2026-05-04 Pod C/D shutdown report 사례.	1.5 min
19	Live demo: <code>thyroid_replay</code> <code>--max-iter 3</code>	<code>cd project/v18_agentic && python3 -m agentic_research.examples.thyroid_replay --max-iter 3</code> · ~1.7s · 5 patterns 모두 exercise · 7 tests pass.	3 min
20	Reproducibility implication + open-source release plan	PyPI release plan · GitHub stars / forks 모니터링 · methods paper venue (Briefings in Bioinformatics / GigaScience / open ABM).	2 min
21	Limitations + generalization caveat	THCA single-project 검증. 다른 prospect biomedical project 에서 validation 필요. Qualitative pattern (정량 metric 부재).	1.5 min
22	Discussion + Q&A		

#	SLIDE TITLE	CONTENT HIGHLIGHTS	TIME
		예상 Q: "AI agent 의 권한 한계?" / "다른 AI platform (GPT-4 / Gemini)?" / "academic ethics?" / "Korean talk 후 영어 venue?"	3 min

25. Talk material files (project/v18_agentic/agentic_research/talk/)

- `agent_vs_agentic_outline_KR.md` – 22-slide outline (01 표의 source)
- `speaker_notes_KR.md` – per-slide script + Q&A
- `demo_cheat_sheet.md` – live-demo script (thyroid_replay)
- `../docs/tutorial_01..05.md` – 5 tutorial walkthroughs
- `../examples/thyroid_replay.py` – end-to-end demo script

25.1 Live demo command (slide 19)

```
$ cd project/v18_agentic
$ python3 -m agentic_research.examples.thyroid_replay --max-iter 3

[Iter 1/3] FixAmplifyAgent → finding fixed: cohort assembly mismatch
[Iter 1/3] TieredParallelOrchestrator → 4 parallel audits done
[Iter 1/3] DualDumpRecorder → memory + log dual-dump
[Iter 1/3] ReviewerLoop → reviewer Q1 simulated, response cached
[Iter 1/3] FailureCatalog → 1 closure battery archived

... (3 iterations, ~1.7s total) ...

$ pytest tests/ -q
..... [100%]
7 passed in 0.41s
```

26. Future plan — Phase A / B / C / D

26.1 Phase A — Korean talk delivery

1. **Korean Bioinformatics Society annual meeting** 일정 + abstract 기한 확인
2. **22-slide deck 제작** — Slides / Keynote에서 outline 기반 (~30 min build)
3. **Pre-talk dry-run** — Yu lab / 협업자 review
4. **Talk delivery** + Q&A 청중 피드백 수집

26.2 Phase B — Methods paper 자체

1. **Methods paper venue 결정** — Briefings in Bioinformatics / GigaScience / open ABM 후보 비교
2. **Reproducibility benchmark 추가** — qualitative pattern 의 정량 metric 정의 (cross-session memory consistency rate / chmod 444 freeze utility / closure battery efficiency)
3. **Co-publish with Paper 6 v52 LODO finding** — Pattern 1 case study + framework

26.3 Phase C — Open-source community

1. **Python pkg PyPI release** — open source license + readme + docs
2. **GitHub repo public** — README + tutorials + contributing guide
3. **Demo screenshots + screencast** — thyroid_replay live demo recording
4. **Community feedback monitoring** — GitHub stars / forks / issues

26.4 Phase D — Cross-project validation (long-term)

1. 다른 prospect biomedical project 에서 framework 적용 (validation)
2. 다른 AI agent platform (GPT-4 / Gemini / Llama) 에서 framework 작동 비교
3. Academic publication ethics committee 와 voice-protection rule 논의

27. Claim boundary table

CLAIM	STATUS
Python package codified (5-pattern + 6-component scaffold)	YES
7/7 unit tests pass	YES
22-slide Korean talk outline ready	YES
5 patterns documented with THCA case study evidence	YES
6 components documented with archetype example	YES
Framework generalization beyond THCA	UNTESTED (single-project)
Quantitative metric for qualitative patterns	PARTIAL (need benchmark)
Korean talk delivered	NO (timeline pending)
Methods paper published	NO (venue TBD)
External lab adoption	NO (Tier 1 future)
Co-publish anchor with Paper 6 (Pattern 1 archetype)	YES (planned)

28. 데이터 / 다운로드

← PAPER 6

v52 LODO finding ·
Pattern 1 archetype

→ PAPER 1

Pattern 2 driver-
exclusion archetype

→ PAPER 2A

Pattern 4 closure
battery archetype

→ PAPER 3

Component B chmod
444 freeze archetype

→ PAPER 2B

Component E
forbidden vocabulary
archetype

V5 MAIN.PDF

489 KB · 20 pages ·
Bioinformatics OUP
draft

V10 AAAI PAPER

301 KB · 11+ pages ·
AAAI 2027 draft

V15 SUBMIT_FINAL.PDF

356 KB · 12 pages ·
Pattern 1 submitted
artifact

UPDATE_NOTE.MD

8K · 8 sections ·
companion MD

← HUB INDEX

8-papers portfolio

Status. Paper 7 = framework codification. v18 agentic_research Python pkg + 7 tests pass + 22-slide Korean talk outline ready. Co-publish with [Paper 6](#) v52 LODO 후보 (Pattern 1 case study + framework).

Live URL. http://40.82.129.113:8012/papers_hub_2026_05_04/paper7.html